



Safir el Tafawok
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Revision One

Part Two: Math

مراجعات سفير التفوق 2025

لتأهيل الراغبين بالالتحاق بمدرسة المتفوقين بعين شمس

و مدارس المتفوقين للعلوم و التكنولوجيا STEM ومدرسة الضبعة النووية

عزيزي الطالب / عزيزتي الطالبة :

تتضمن هذه الاسئلة أسئلة متعددة الأفكار , قد تجد بعضها سهلا و البعض الآخر صعبا ,
حاول الإجابة علي جميع الأسئلة .



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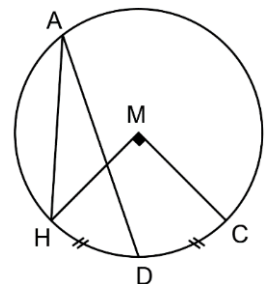
1- In the opposite figure:

If the measure of arc (DC) is equal to the measure of arc (DH), and the measure of angle (HMC) is 90 degrees, then the measure of angle (HAD)

- 30
- 11.25
- 45
- 22.5



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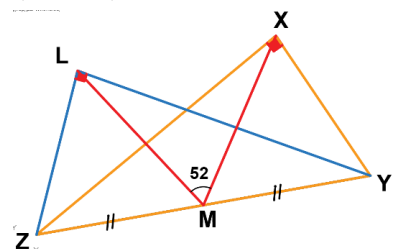
2- In the adjacent figure:

If $m(X) = m(L) = 90^\circ$, $YM = MZ$, $m(XML) = 52^\circ$, then $m(XZL) = \dots\dots\dots^\circ$

- 26
- 32
- 64
- 52



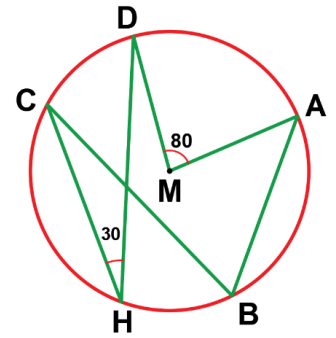
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3- In the adjacent figure:

$m(\angle ABC) = \dots\dots\dots$ degrees.

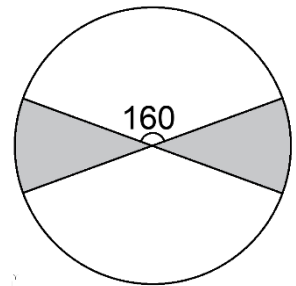
- ☐ 40
- ☐ 50
- ☐ 60
- ☐ 70



4- Look at the following shape , A circle of center M and radius = 6 .

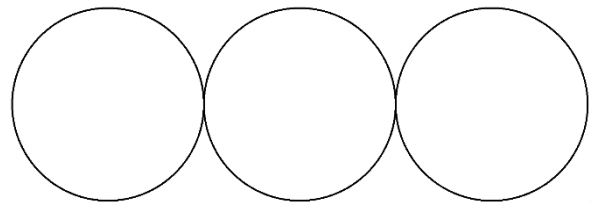
What is the sum of the areas of the two shaded figures ?

- ☐ 4π
- ☐ 4.5π
- ☐ 6π
- ☐ 7.5π
- ☐ 10π



5- Number of Axis of symmetry in the figure =....., Number of external tangents =.....

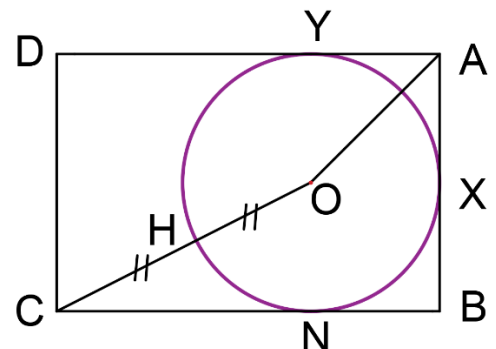
- ☐ 2 , 2
- ☐ 3 , 3
- ☐ 4 , 4
- ☐ 2 , 4



6- In the opposite figure:

A B C D rectangle, the circle O touches AB , BC , AD at the points X, B, Y respectively, if $OH = HC$; Then $(\angle AON) = \dots\dots\dots^\circ$

- ☐ 165
- ☐ 135
- ☐ 120
- ☐ 150

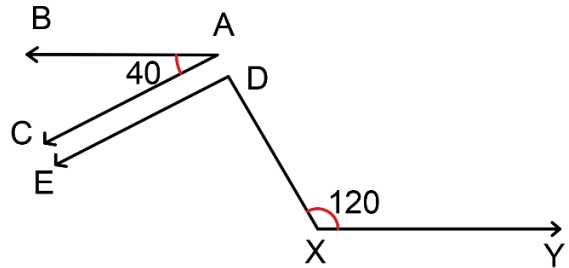


7-In the opposite figure:

$AB \parallel XY$, $AC \parallel DE$, $m(\angle A) = 40^\circ$, $m(\angle X) = 120^\circ$,

Then $m(\angle XDE) = \dots\dots\dots$

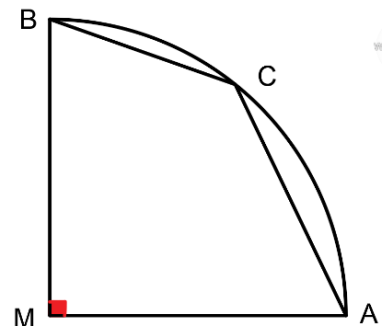
- ☐ 75°
- ☐ 60°
- ☐ 80°
- ☐ 70°



8- In the opposite figure:

Quarter of a circle M, $m(\angle ACB) = \dots\dots\dots^\circ$

- ☐ 120
- ☐ 130
- ☐ 125
- ☐ 135

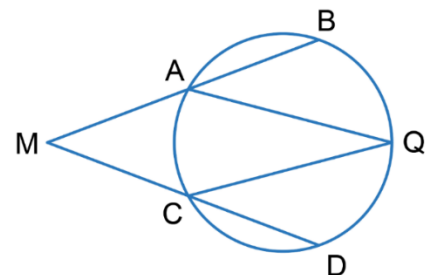


9- In the opposite figure:

Point A, B, Q, D, and C lie on the shown circle and $m(\widehat{BQ}) = 42^\circ$,

$m(\widehat{QD}) = 38^\circ$. Then $m(\angle Q) + m(\angle M) = \dots\dots\dots$

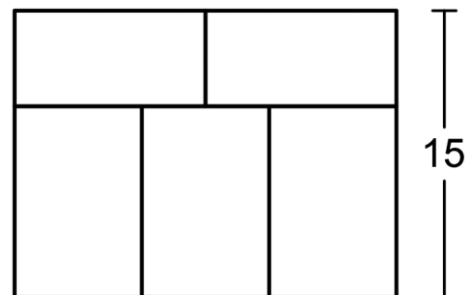
- ☐ 40
- ☐ 80
- ☐ 62
- ☐ 46



10- In the adjacent figure

There are five identical rectangles arranged to form a larger rectangle with a width of 15 cm. The area of the resulting rectangle is = $\dots\dots\dots \text{cm}^2$

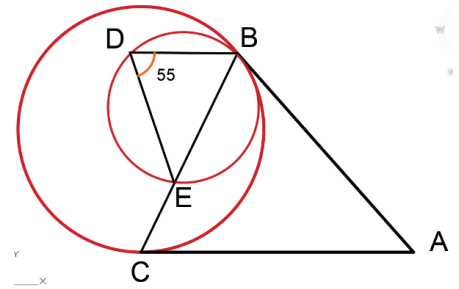
- ☐ 330
- ☐ 270
- ☐ 360
- ☐ 300



11- In the opposite figure:

Two internally tangent circles at point B, if AB and AC are two tangents, then $m(\angle A) = \dots\dots\dots^\circ$

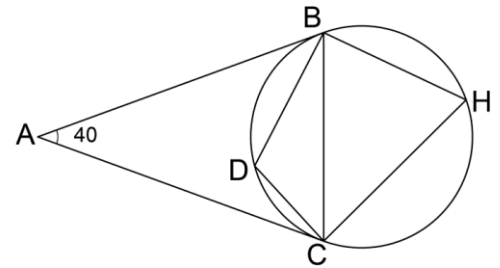
- ☐ 60
- ☐ 75
- ☐ 55
- ☐ 70



12- In the given diagram:

A, B, and C are tangents to the circle at points B and C, with $\angle(BAC) = 40^\circ$. D and C are on the circle. If the ratio $\angle(BD) : \angle(DC)$ is 3:2, then $\angle(ABD) = \dots\dots\dots^\circ$

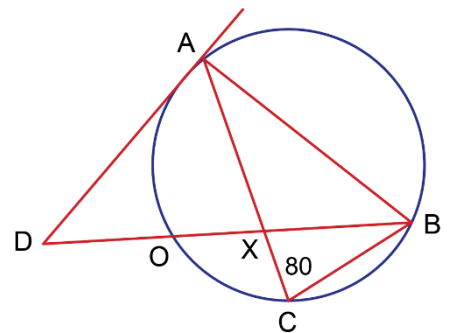
- ☐ 28
- ☐ 35
- ☐ 42
- ☐ 56



13- In the opposite figure:

AD is a tangent to the circle at A, and $AB = AD$, then the measure of angle (D) is $\dots\dots\dots$ degrees.

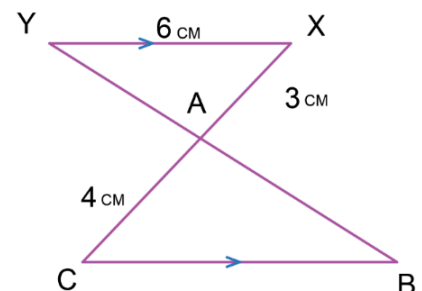
- ☐ 40
- ☐ 45
- ☐ 50
- ☐ 60



14- In the given figure:

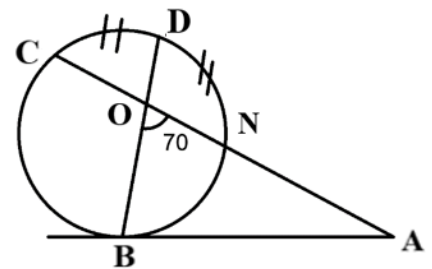
If $\triangle ABC \sim \triangle AYX$, then the length of BC = $\dots\dots\dots$ cm

- ☐ 4
- ☐ 6
- ☐ 8
- ☐ 10



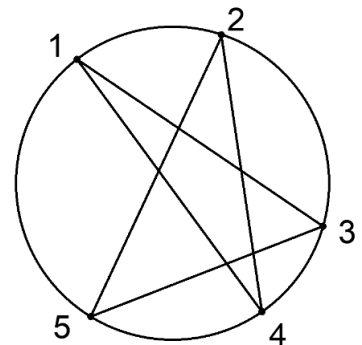
15- In the opposite figure : AB is a Tangent to a Circle in B , if (DN) = DC, Then the measurement of the angle (ABO) =

- 45
- 40
- 90
- 70



16- Find the sum of angles 1,2,3,4,5 in the following star shape in degrees

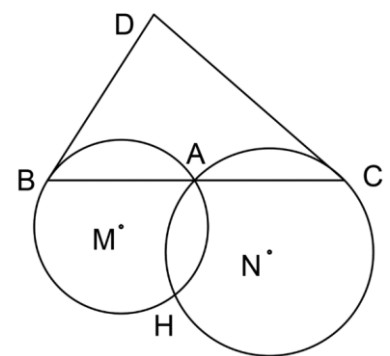
- 90
- 180
- 270
- 360
- 300



17- In the opposite figure:

A circle $M \cap N = \{A, B\}$. $\overline{BD}$ is a tangent to the circle M. $\overline{CD}$ is a tangent to the circle N. $m(\text{minor}\widehat{AB}) = (x + 2y)^\circ$, $m(\text{minor}\widehat{AC}) = (3x + 2y)^\circ$ where $x + y = 50^\circ$. Then $m(\angle D) = \dots\dots\dots^\circ$

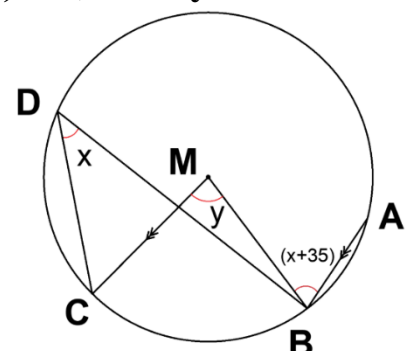
- 60
- 70
- 80
- 90



18- In the opposite figure:

If $AB \parallel MC$, $m(\angle ABM) = (x + 35)^\circ$, $m(\angle M) = y^\circ$, $m(\angle D) = x^\circ$, then $y = \dots\dots\dots$

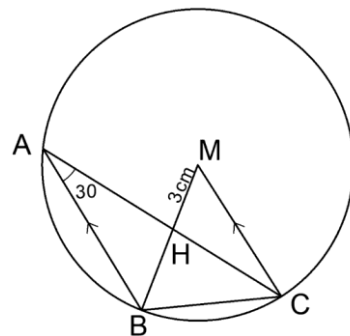
- 50
- 70
- 60
- 80



19- In the opposite figure:

AB, AC are two chords in circle M . If $m(\angle A) = 30$, and $MH = 3$ cm, then the circumference of circle $M = \dots\dots\dots \pi$ cm.

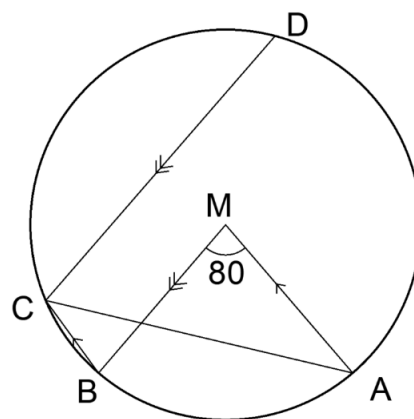
- ☐ 15
- ☐ 12
- ☐ 9
- ☐ 6



20 -In the opposite figure:

If $AM \parallel BC$, $MB \parallel DC$, $m(\angle M) = 80$, then $m(\angle ACD) = \dots\dots\dots$

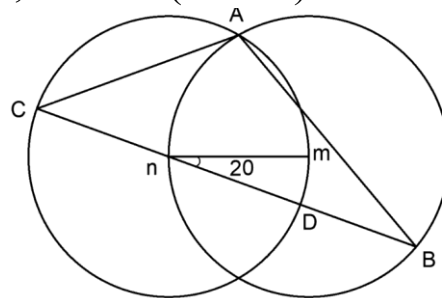
- ☐ 50
- ☐ 70
- ☐ 80
- ☐ 60



21- In the opposite figure:

M, N are two congruent circles. If $m(\angle MNB) = 20$, then $m(\angle BAC) = \dots\dots\dots$

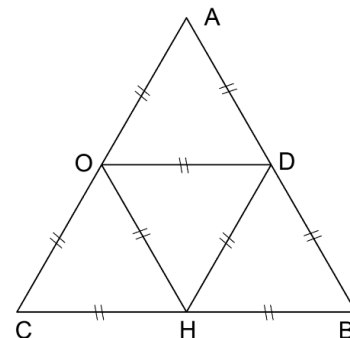
- ☐ 120
- ☐ 110
- ☐ 100
- ☐ 90



22 -In the opposite figure:

$\triangle ABC$ is equilateral. If we denote the number of congruent triangles by the symbol X , and denote the number of similar triangles by the symbol Y , then $X:Y = \dots\dots\dots$

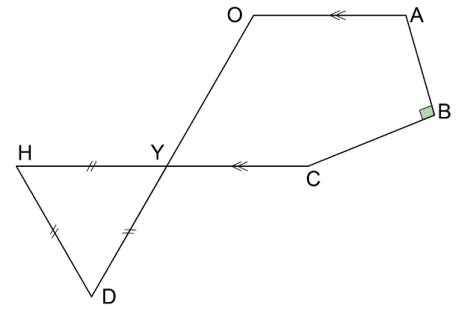
- ☐ 2:1
- ☐ 5:4
- ☐ 4:5
- ☐ 1:1



23- In the opposite figure:

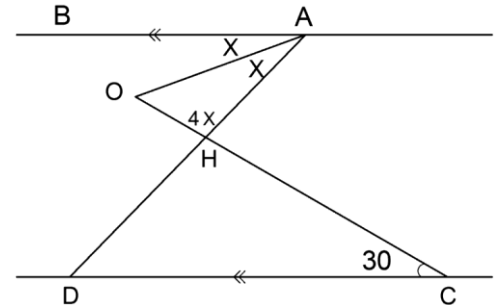
If $M(\angle A) : M(\angle C) = 2 : 3$, then $M(\angle A) = \dots\dots\dots$

- 128
- 112
- 162
- 108



24- If $AB \parallel CD$, then $(\angle BAO) = (\angle OAH) = X$, and $(\angle AHO) = 4X$
 $(\angle HCD) = 30^\circ$ Then $X = \dots\dots\dots$

- 10
- 25
- 30
- 40



25- The Perimeter of a circle with an arc of 16 cm and a measure of 144° is cm

- 20
- 40
- 40π
- 20π

26- A metal ball with a diameter of 12 cm. If it melts and turns into a right circular cylinder with a radius of 6 cm. What is the height of the cylinder?

- 4 cm
- 3 cm
- 6 cm
- 8 cm



27- If $(1 + x)(1 + x^2)(1 + x^4) = 1$, then the set of solutions in $\mathbb{R}$ is

- $\{0,1\}$
- $\{0\}$
- $\{1\}$
- $\{0,1,7\}$



28- The solution set of the equation: $x - \sqrt{x} = 2$ in $\mathbb{R}$ is ...

- ☐ {1,4}
- ☐ {1}
- ☐ {4}
- ☐ $\emptyset$



29- If $X + \frac{1}{X} = 5$, then the value of $\frac{2X}{3X^2 - 5X + 3} = \dots$

- ☐ $\frac{3}{10}$
- ☐ $\frac{1}{5}$
- ☐ $\frac{2}{5}$
- ☐ $\frac{1}{10}$

30- If $X + \frac{1}{X} = 7$, then the value of $\frac{2X}{3X^2 - 5X + 3} = \dots$

- ☐ $1/8$
- ☐ $1/16$
- ☐ $1/7$
- ☐ $3/16$



31- If $\frac{97}{19} = a + \frac{1}{b + \frac{1}{c}}$, where a, b, c are all positive integers,

then $a + b + c = \dots\dots\dots$

- ☐ 17
- ☐ 18
- ☐ 19
- ☐ 16



32- $X + 1 < \sqrt[3]{49} < X + 2$, where X is a positive integer, then $\frac{x+1}{x} = \dots\dots\dots$

- ☐ $3/2$
- ☐ $4/3$
- ☐ 1
- ☐ $3/4$



33 If $\frac{5}{7} = \frac{a}{b}$, and $\frac{c}{d} = \frac{2a}{3b}$, then $\frac{ac}{db} = \dots\dots\dots$

- ☐ 163 /56
- ☐ 56 /163
- ☐ 147 /50
- ☐ 50 /147

34- A camp with 700 men lasts for 30 days, if the man's share of food is 2.5 kg per day. If 100 men leave the camp, how many days will the camp last if the man consumes 3.5 kg of food per day?

- ☐ 20
- ☐ 15
- ☐ 18
- ☐ 25



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35- If L , 3 , 6 , M are in serial proportion , Then $2L - M = \dots\dots\dots$

- ☐ 9
- ☐ -9
- ☐ 15
- ☐ -12

36- If B is an average between $(B+2)$, $(B-6)$. Then $B = \dots\dots\dots$

- ☐ 3-
- ☐ 6
- ☐ 3
- ☐ 6-



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37- A bag contains a certain number of toy cubes with the letters A, B, C, D written on them, such that the ratios of the cubes are $A : B : C : D = 4 : 7 : 3 : 1$. If the number of cubes with letter (A) is 50 more than the number of cubes with letter (C), then the number of cubes with letter (B) is

- ☐ 350
- ☐ 250
- ☐ 200
- ☐ 300



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38- If the price of a commodity increases by 20% and then decreases by 20% from the new price, then the price is

- ☐ Decrease by 4%
- ☐ Increase by 4%
- ☐ Increase by 0%
- ☐ Increase by 40%



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39- If $a + b : b + c : c + a = 6 : 7 : 8$; and $a + b + c = 14$; then $c =$

- ☐ 4
- ☐ 6
- ☐ 3
- ☐ 2

40- If $(A + B) : (B + C) : (C + A) = 6 : 7 : 8$; Then $A : B : C =$

- ☐ 7 : 8 : 9
- ☐ 4 : 5 : 7
- ☐ 5 : 7 : 9
- ☐ 7 : 5 : 9



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41- A computer club with 60 members and 40% of the members were girls. If 20 boys later join the club, then the percentage of girls now is

- ☐ 30%
- ☐ 50%
- ☐ 70%
- ☐ 60%

42- If A, B, C, and D are positive quantities in a series proportion, and $C = 9$ $A = 9$, then $B + D$ =

- ☐ 27
- ☐ 36
- ☐ 81
- ☐ 30



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43- If m , 5, n , and 20 are in a serial ratio, then the positive values of m and n are

- ☐ 10, 20
- ☐ 5, 10
- ☐ 2.5, 10
- ☐ 2.5, 20



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44- When dividing a positive integer n by 8, the remainder is 7, when dividing $5n$ by 8, the modulus is

- ☐ 4
- ☐ 5
- ☐ 3
- ☐ 6

45- When dividing the number k by 7 the remainder is 2, so the remainder of dividing the number $4k$ by 7 is

- ☐ 3
- ☐ 1
- ☐ 6
- ☐ 2



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46- Dividing a number by (374) The remainder is (39) , Then Dividing this number by (17) , The remainder =.....

- ☐ 11
- ☐ Zero
- ☐ 3
- ☐ 5

47- A two-digit number has 5 tens. If the position of its digits is reversed, the resulting number is 36 more than the original number. The ones digit of the original number..... =

- ☐ 7
- ☐ 3
- ☐ 9
- ☐ 5



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48- A two-digits number its ones digit is X and its tens digit is Y, if the product of its digits is 18, and the ratio between the number to the sum of its digits is 4, then $X^2 - Y^2 = \dots\dots\dots$

- ☐ 12
- ☐ 45
- ☐ 6
- ☐ 27

49- Number composed of two digits, units digit is 'X', and tens digit is 'Y' The product of its digits equals 8, and the ratio of the number to the sum of its digits is 4. Then, $X^2 - Y^2 = \dots\dots\dots$

- ☐ 2
- ☐ 12
- ☐ 16
- ☐ 48



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50- A positive two-digit integer composed of two digits. If its digits are reversed, the resulting number is 54 more than the original number. The absolute difference between the digits of the original number is $\dots\dots\dots$

- ☐ 7
- ☐ 6
- ☐ 8
- ☐ 5



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51- If the ratio between Ahmed's age and Nada's age now is 6:5, and after 15 years this ratio will become 9:8, then Nada's age now is...

- ☐ 30 years
- ☐ 6 years
- ☐ 25 years
- ☐ 5 years

52- The arithmetic mean for the value 5, 4, 3, 2 is $\dots\dots\dots$

Select one:

- ☐ 3.5
- ☐ 5
- ☐ 4
- ☐ 4.5



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53- The true statement for the data:

1, -1 , 0 , 2 , 3 , 4 , 4 , 6 , 8 , 10 , 11 is

- ☐ **Mode = median = 4**
- ☐ **Mean = median = 4**
- ☐ **Mean = mode = median**
- ☐ **Mean = mode = 4**



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54- If the median of a set of numbers is 7 and the larger number increases by 3; The median for the new values is

- ☐ **4**
- ☐ **7**
- ☐ **5**
- ☐ **10**

55- Ahmad calculated the arithmetic mean for five numbers and found it to be 7, but then discovered that he mistakenly entered 11 instead of 6 in one of the inputs. What is the correct arithmetic mean?

- ☐ **9**
- ☐ **7.5**
- ☐ **8.5**
- ☐ **8**

56- A set of values has a sum of 63, and its median rank is 4. Therefore, its arithmetic mean equals:

- ☐ **7**
- ☐ **21**
- ☐ **14**
- ☐ **9**



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57- If the arithmetic mean of the values: A , A + 9, A + 1, A + 7, A + 3 equals X, and their median equals Y, then if X : Y = 3 : 2, then A can equal

- ☐ **1**
- ☐ **9**
- ☐ **7**
- ☐ **3**

58 -If the mean of the numbers: 11, 6, k, 7, k, k equals the mode of these numbers, then k equals.....

- ☐ 5
- ☐ 7
- ☐ 8
- ☐ 6

59- A, B, C, D , and E are five numbers whose arithmetic mean is 5, and the arithmetic mean of the numbers A, B , and C is 3, then the arithmetic mean of the two numbers D and E is ...

- ☐ 16
- ☐ 2
- ☐ 4
- ☐ 8



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60- Several values add up to 88, and their median rank is 6. Therefore, their arithmetic mean is..... =

- ☐ 11
- ☐ 8
- ☐ 9
- ☐ 7



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61- If the median of a set of numbers is 9 and the largest number increases by 5, then the median of the new values is equal to...

- ☐ 45
- ☐ 7
- ☐ 14
- ☐ 1

62- Huda painted the faces of the cube either in red or yellow; If the probability of yellow color appearing on the upper face when thrown in a random way is equal to $1/3$. How many faces are colored red?

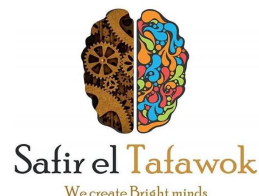
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 1



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63- There are 16 glass ball in a bag : 8 red and 8 blue , Sara chose a random ball and it was a red ball , Sara put the ball back in the bag . What is the probability to pick a red ball ?

- ☐ 4 / 10
- ☐ 1 / 5
- ☐ 1 / 10
- ☐ 1 / 2



64- If a, b are two events such that $L(a \cup b) + L(a \cap b) = 7/8$, $L(a) = 2L(b)$; Then for $L(b) = \dots\dots\dots$

- ☐ 5/12
- ☐ 17/24
- ☐ 7/24
- ☐ 7/12



65- A group of 150 tourists from Japan , 10 of them neither speaks English nor Deutsch , 120 of them speaks Deutsch and 130 tourists speaks English . How many tourists that can speak both languages?

- ☐ 10
- ☐ 20
- ☐ 110
- ☐ 140
- ☐ 100

66- There are two boxes A and B . The number of red balls in each box is equal, and the number of white balls in box B exceeds the number of white balls in box A by one. If the probability of drawing a red ball from box A is x and the probability of drawing a red ball from box B is y , then the number of red balls in each box equals:

- ☐ $\frac{xy}{x-y}$
- ☐ $\frac{1}{xy}$
- ☐ $\frac{xy}{x+y}$
- ☐ $\frac{1}{x} - \frac{1}{y}$



67- A bag contains a number of cards, $\frac{1}{6}$ of them are green, $\frac{1}{12}$ of them are yellow, $\frac{1}{2}$ of them are white and $\frac{1}{4}$ of them are blue.

If someone chose a card from the bag randomly, then the color which has the greatest probability is

- ☐ Blue
- ☐ Green
- ☐ White
- ☐ Yellow



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68- In the experiment of forming a number from two different digits from the set of numbers 1, 2, 3, the probability that the number is even..... .

- ☐ $\frac{2}{3}$
- ☐ $\frac{4}{9}$
- ☐ $\frac{2}{9}$
- ☐ $\frac{1}{3}$

69- If the probability of guessing the correct answer to a question is $\frac{x}{2}$, and the probability of not guessing the correct answer to this question is $\frac{2}{3}$, then $x = \dots$

- ☐ $\frac{3}{4}$
- ☐ $\frac{2}{3}$
- ☐ $\frac{1}{2}$
- ☐ $\frac{1}{3}$



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70- If A and B are two events in a sample space for a random experiment where $A \subset B$, $P(A) = 0.2$, $P(B) = 0.6$, then $P(B - A) = \dots\dots\dots$

- ☐ $\frac{2}{5}$
- ☐ $\frac{3}{5}$
- ☐ $\frac{1}{5}$
- ☐ $\frac{4}{5}$

71- The A and B are two events in a sample space for a random experiment where $A \subset B$, $P(A) = 0.3$, $P(B) = 0.9$,

Then $(B - A) = \dots\dots\dots$

- ☐ $\frac{4}{5}$
- ☐ $\frac{1}{5}$
- ☐ $\frac{3}{5}$
- ☐ $\frac{2}{5}$



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72- If X and Y are two events in a sample space for a random experiment, $P(Y) = \frac{2}{5}$, $P(X) = P(X')$, $P(X \cap Y) = \frac{1}{5}$, then: $P(X \cup Y) = \dots$

- ☐ $\frac{9}{10}$
- ☐ $\frac{7}{10}$
- ☐ $\frac{3}{5}$
- ☐ $\frac{1}{2}$



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73- There are 25 cards numbered from 1 to 25. One card is drawn at random. What is the probability that the number on this card is not divisible by 3?

- ☐ $\frac{17}{25}$
- ☐ $\frac{8}{25}$
- ☐ $\frac{19}{25}$
- ☐ $\frac{6}{25}$



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74- A container contains 36 colored beads all of the same size:
Some of them is blue, some green, some red, and the rest is yellow.
A bead is drawn from the container without looking.
If the probability that it is blue is $\frac{4}{9}$, then the number of blue beads equals
..... bead.

- ☐ 4
- ☐ 8
- ☐ 16
- ☐ 18



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75- A box contains 5 oranges, 6 oranges, and X bananas. If the probability of choosing an orange from the event is $\frac{1}{3}$, then $X = \dots\dots\dots$
Choose one of the options

- ☐ 7
- ☐ 5
- ☐ 6
- ☐ 8



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